Simulation Output Summary

Integrated Energy Systems Optimiser (IESO) v24.03 · www.github.com/greoux-research/ieso

March 25, 2024 09:24 AM

Demand

Demand for Electricity:

- Annual Demand in MWh: 50000000
- Hourly Profile: Figure 1
- Non-Service Penalty in $ per MWh: 20000
- Supply Sources: Power Grid

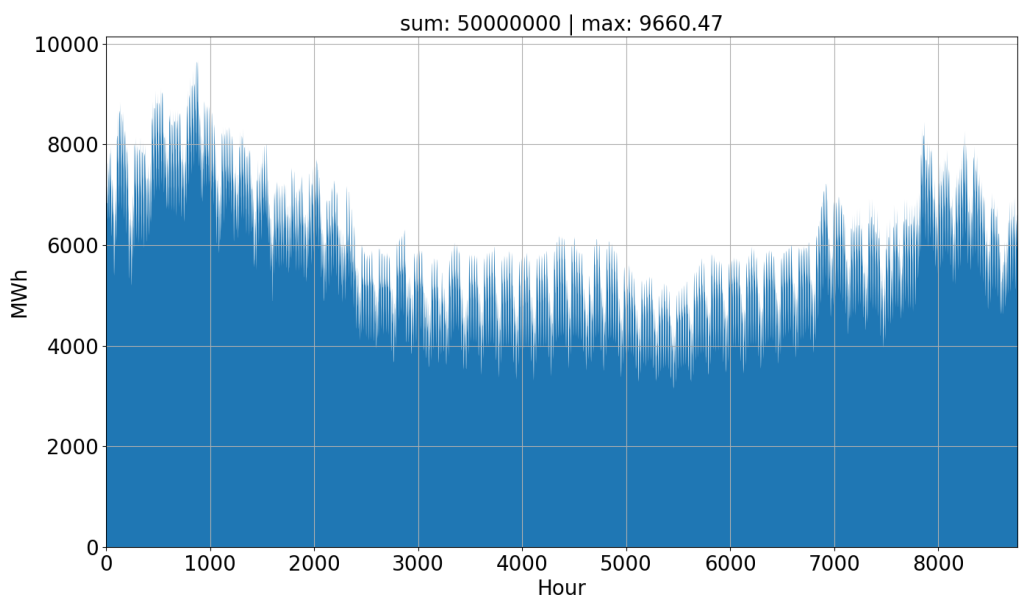

Figure 1: Demand for Electricity

Generators

Generator "nucl":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 456466
- Variable Costs in $ per MWh: 11.5
- Variable Emissions in $ per MWh: 0
- The optimiser did not suggest adding generation capacity

Generator "coal":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 222392
- Variable Costs in $ per MWh: 26.84
- Variable Emissions in $ per MWh: 880
- Capacity in MW: 4281.41
- Electricity Generation: 31441178 MWh (Figure 2)
- Number of Successive Changes in Output: 788

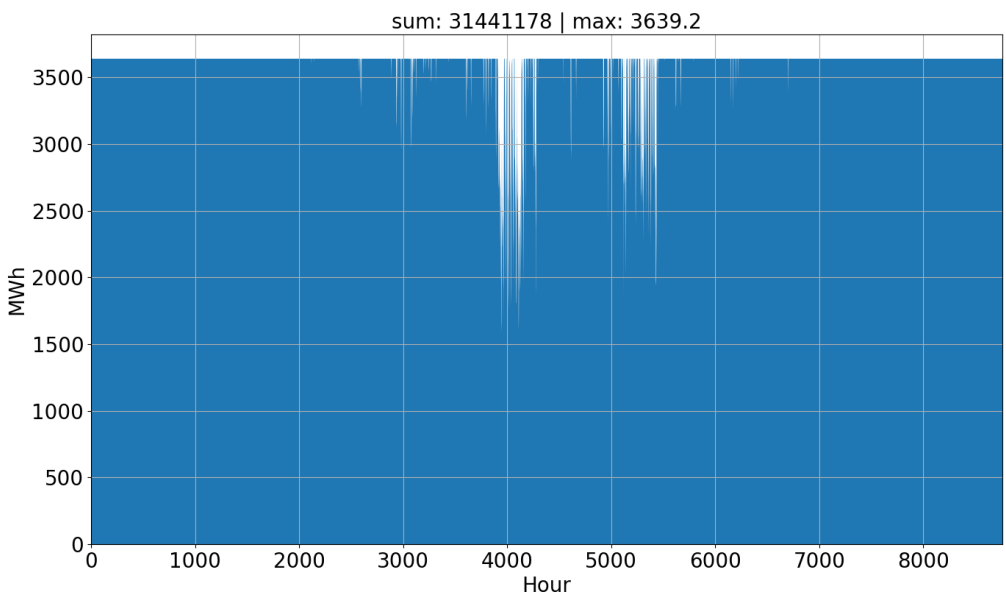

Figure 2: Electricity Generation by "coal"

Generator "ccgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 98749
- Variable Costs in $ per MWh: 56.44
- Variable Emissions in $ per MWh: 365
- Capacity in MW: 1392.09
- Electricity Generation: 3227254 MWh (Figure 3)

- Number of Successive Changes in Output: 1559

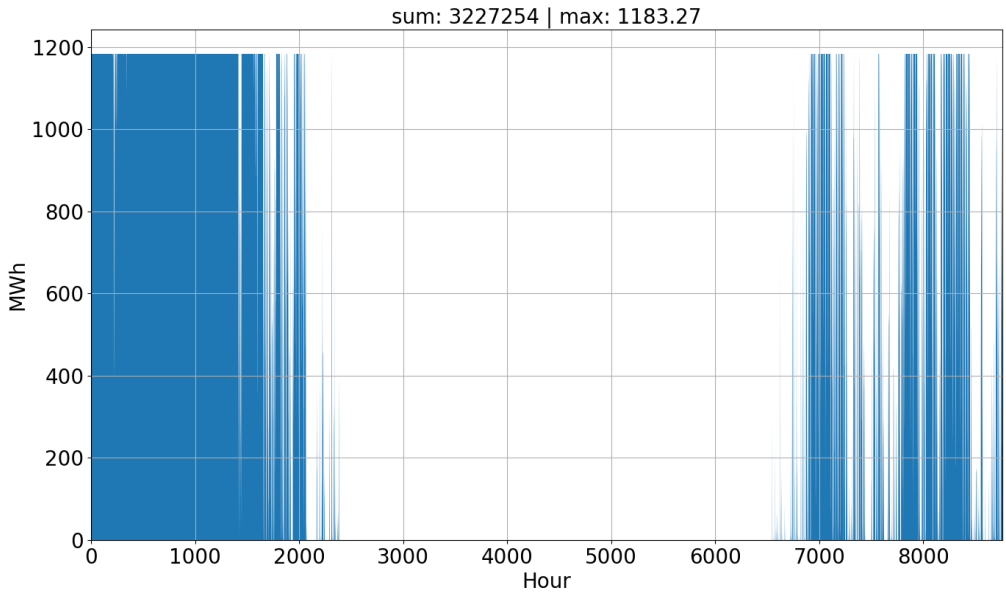

Figure 3: Electricity Generation by "ccgt"

Generator "ocgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 75182
- Variable Costs in $ per MWh: 96.11
- Variable Emissions in $ per MWh: 523
- Capacity in MW: 1413.61
- Electricity Generation: 158365 MWh (Figure 4)
- Number of Successive Changes in Output: 380

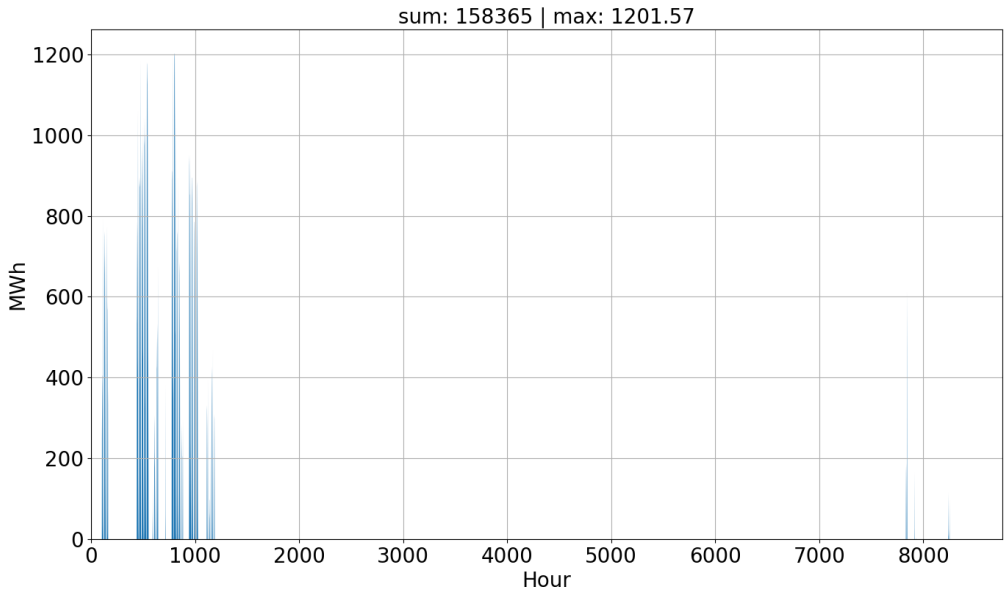

Figure 4: Electricity Generation by "ocgt"

Generator "solr":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 70046
- Variable Costs in $ per MWh: 3e-06
- Variable Emissions in $ per MWh: 0
- The optimiser did not suggest adding generation capacity

Generator "wind":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 133553
- Variable Costs in $ per MWh: 6e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 1935.55
- Electricity Generation: 5086633 MWh (Figure 5)
- Curtailment: 0 Hours; 0 MWh or 0.0 % of the Tech.'s Annual Gen. Volume

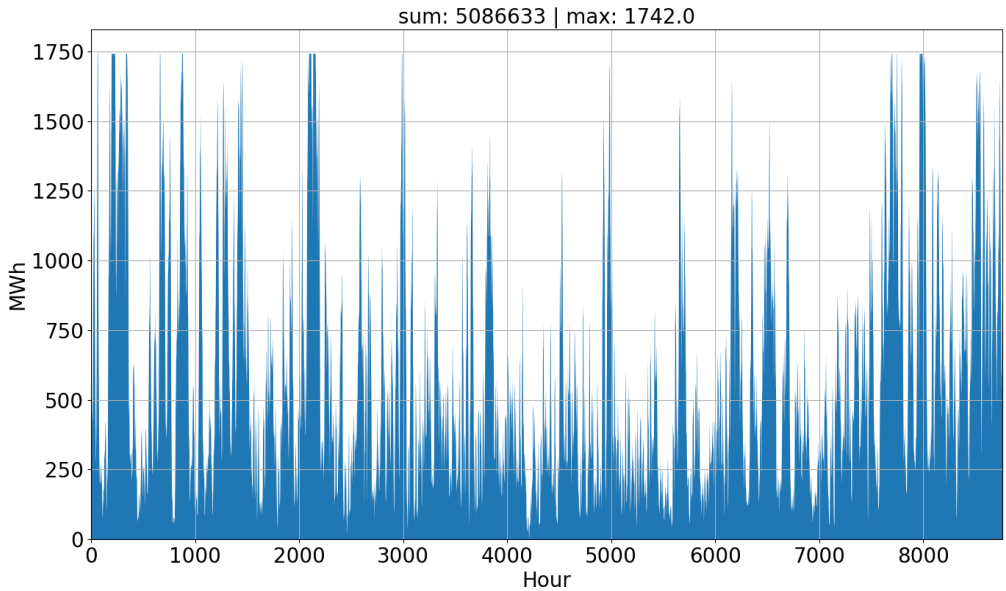

Figure 5: Electricity Generation by "wind"

Generator "hror":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 0
- Variable Costs in $ per MWh: 9e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 1000.0
- Electricity Generation: 5035207 MWh (Figure 6)
- Curtailment: 0 Hours; 0 MWh or 0.0 % of the Tech.'s Annual Gen. Volume

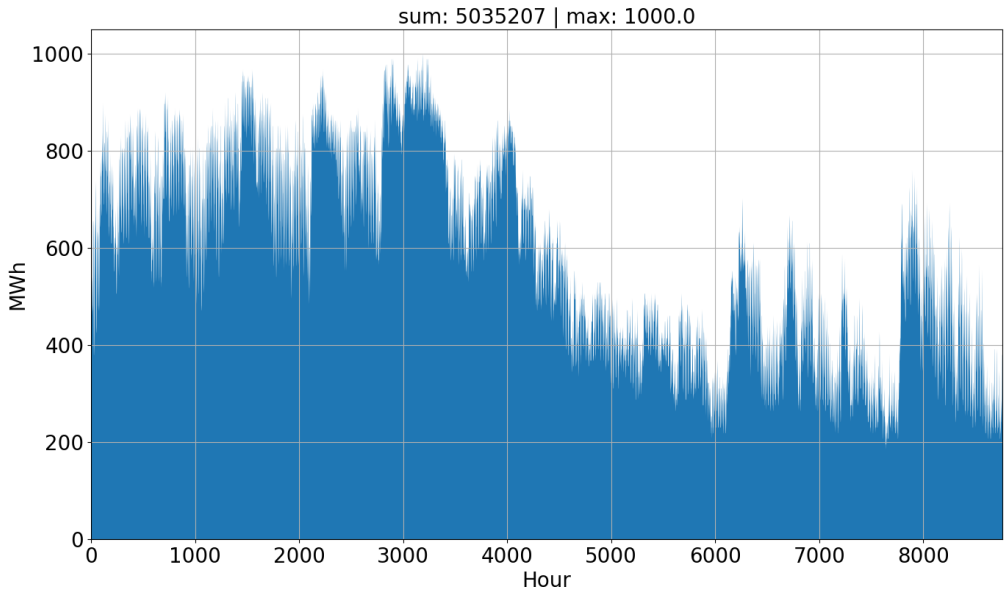

Figure 6: Electricity Generation by "hror"

Flexibility Means

Flexibility Mean "bstr":

- Type: Electricity Storage System
- Fixed Costs in $ per MWh per Year: 27248
- Hours of Storage at Maximum Discharge: 4
- Round Trip Efficiency: 0.85
- The optimiser did not suggest adding storage capacity

Flexibility Mean "hstr":

- Type: Electricity Storage System
- Fixed Costs in $ per MWh per Year: 0
- Hours of Storage at Maximum Discharge: 730
- Round Trip Efficiency: 0.85
- Capacity in MWh: 730000.0
- Electricity Charged (-) and Discharged (+): Figure 7
- Electricity Stored: Figure 8

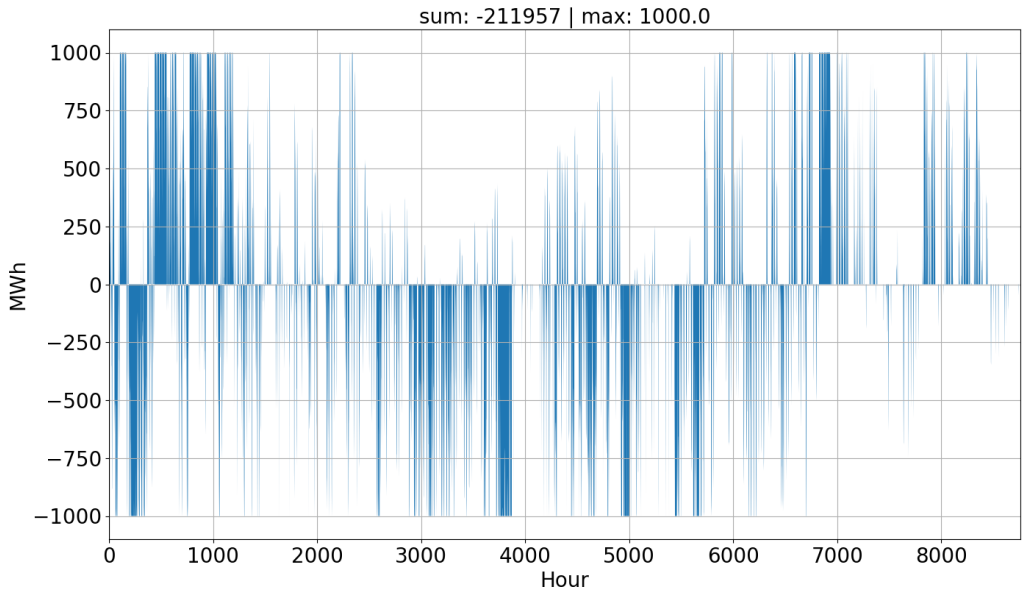

Figure 7: Electricity Charged (-) and Discharged (+) by "hstr"

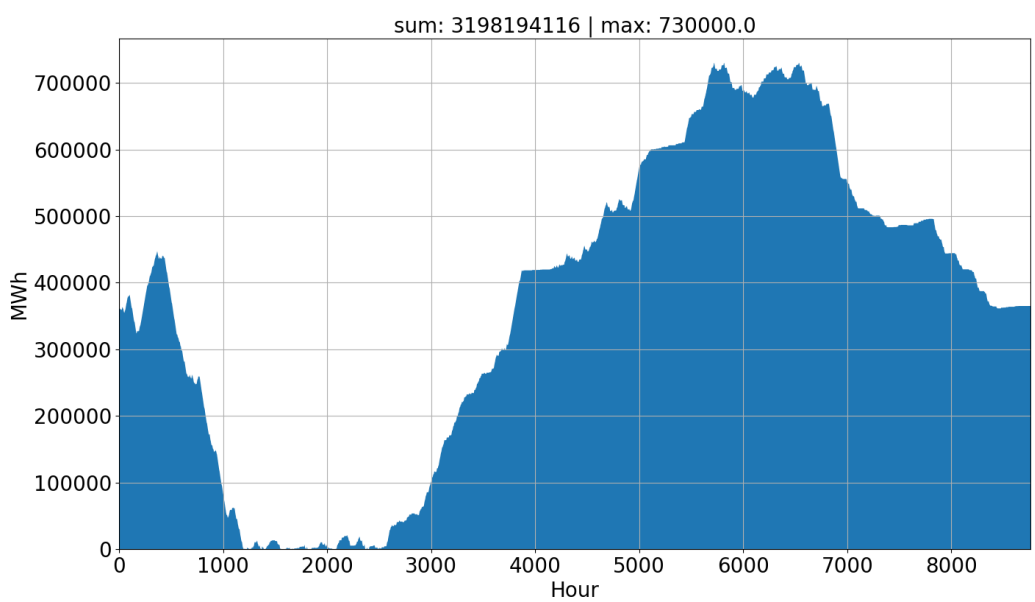

Figure 8: Electricity Stored by "hstr"

Flexibility Mean "hdam":

- Type: Hydroelectric Dam

- Fixed Costs in $ per MWh per Year: 0
- Hours of Storage at Maximum Discharge: 730
- Round Trip Efficiency: 1
- Capacity in MWh: 730000.0
- Electricity Charged (-) and Discharged (+): Figure 9
- Electricity Stored: Figure 10

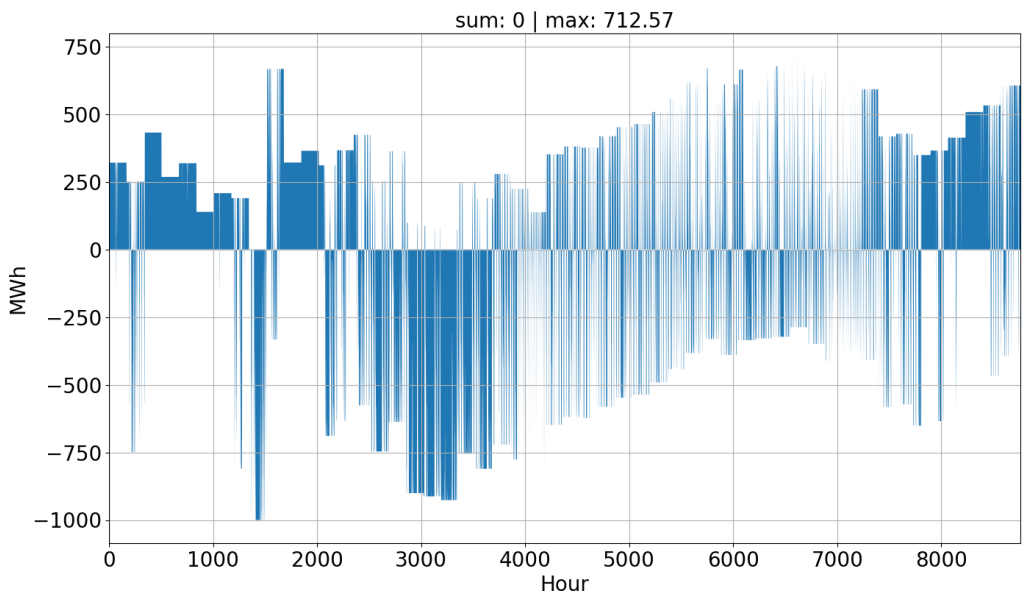

Figure 9: Electricity Charged (-) and Discharged (+) by "hdam"

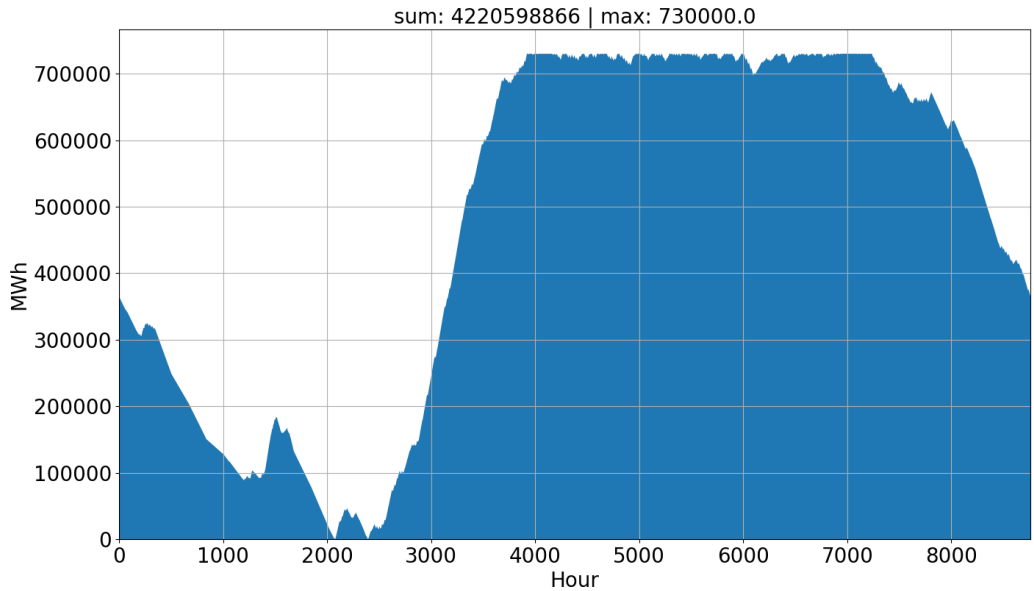

Figure 10: Electricity Stored by "hdam"

Costs and Emissions

Generation Costs and Emissions on a per MWh Basis:

- Generation Costs in $ per MWh: 55.52
- Generation Emissions in kg per MWh: 643.6

Prices

Electricity:

- Average Price in $ per MWh: 65.97
- Price Profile: Figure 11

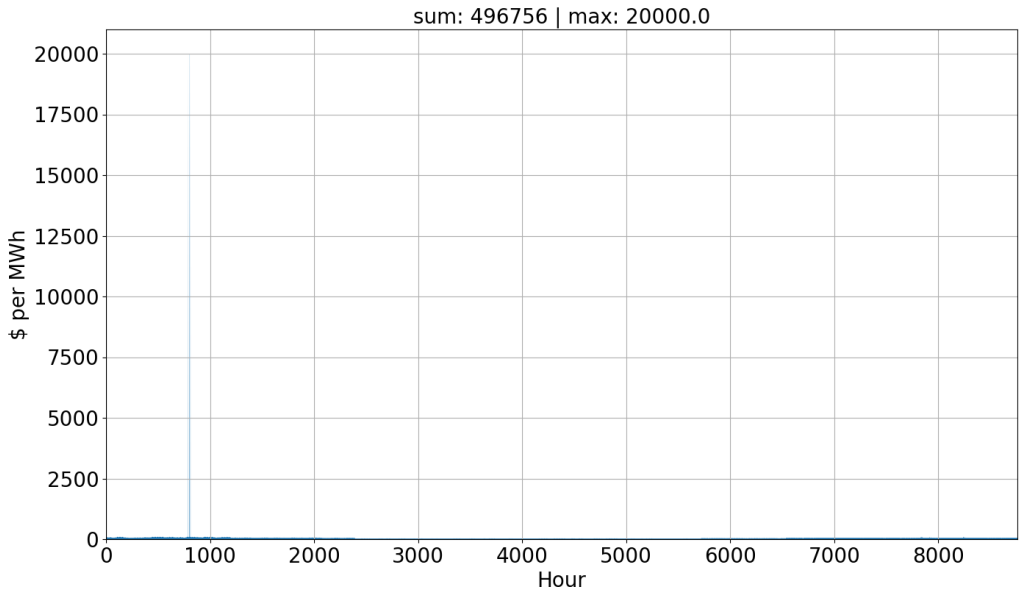

Figure 11: Electricity Price Profile

Autogenerated by Integrated Energy Systems Optimiser (IESO) v24.03 on March 25, 2024 09:24 AM